

LAB EXPERIMENTS

Tool: Python & R Programming

Register No: 192424189

Name: Dhanusha D

Date: 22-07-2026

Subject: Data Handling & Visualization

Code: DSA0604

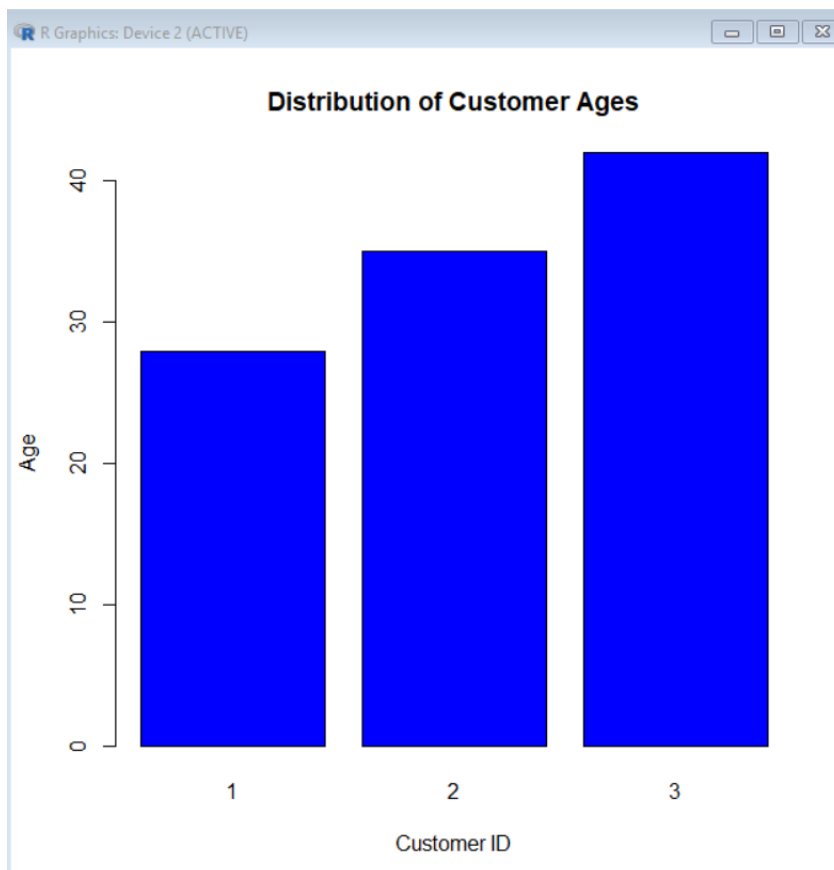
SET 16: CUSTOMER DEMOGRAPHICS ANALYSIS

1. Bar Chart of Customer Ages

CODE:

```
barplot(customer$Age,  
        names.arg = customer$Customer_ID,  
        col = "blue",  
        xlab = "Customer ID",  
        ylab = "Age",  
        main = "Distribution of Customer Ages")
```

OUTPUT:

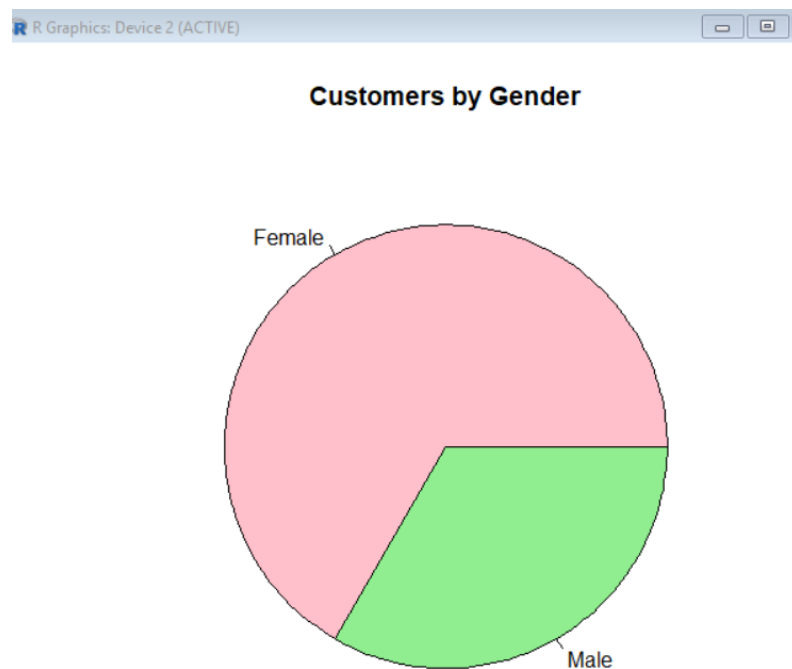


2. Pie Chart for Gender Distribution

CODE:

```
gender <- table(customer$Gender)
pie(gender,
    col = c("pink", "lightgreen"),
    main = "Customers by Gender")
```

OUTPUT:



3. Display Demographic Table

CODE:

```
Customer <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
print(Customer)
```

OUTPUT:

```
> customer
  Customer_ID Age Gender Income
1           1  28 Female  50000
2           2  35  Male  60000
3           3  42 Female  75000
> |
```

4. Tableau Dashboard

CODE:

```
library(shiny)

library(plotly)

customer <- data.frame(

  CustomerID = c(1,2,3),

  Age = c(28,35,42),

  Gender = c("Female","Male","Female"),

  Income = c(50000,60000,75000)

)

gender_count <- as.data.frame(table(customer$Gender))

colnames(gender_count) <- c("Gender","Count")

ui <- fluidPage(

  titlePanel("Customer Demographics Dashboard"),

  fluidRow(

    column(6, plotlyOutput("barChart")),

    column(6, plotlyOutput("pieChart"))

  ),

  fluidRow(

    tableOutput("customerTable")

  )

)

server <- function(input, output){

  output$barChart <- renderPlotly({

    plot_ly(
```

```

customer,

x = ~CustomerID,

y = ~Age,

type = "bar"

)

})

output$pieChart <- renderPlotly({

  plot_ly(

    gender_count,

    labels = ~Gender,

    values = ~Count,

    type = "pie"

  )

})

output$customerTable <- renderTable({

  customer

})

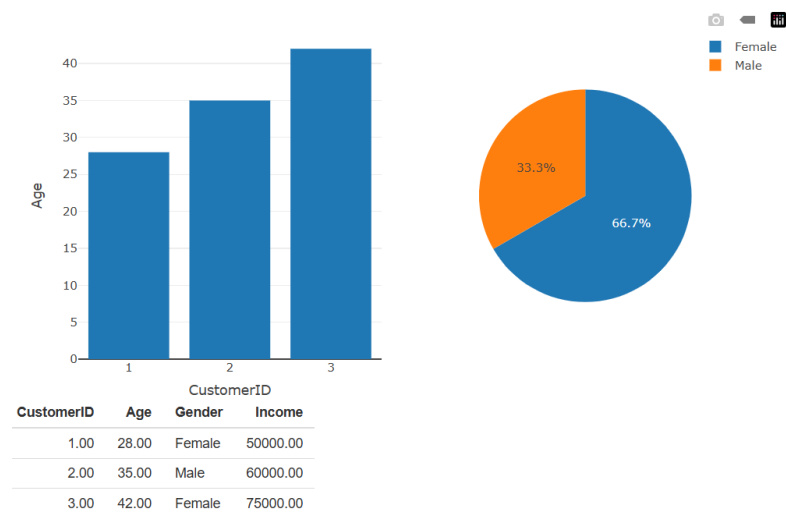
}

shinyApp(ui, server)

```

OUTPUT:

Customer Demographics Dashboard



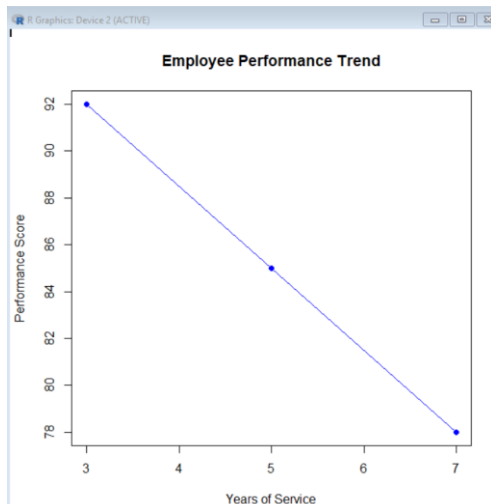
SET 17: EMPLOYEE PERFORMANCE ANALYSIS

1. Line Chart to Visualize the Performance Trend of Employees Over Time

CODE:

```
Employee <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
print(Employee)  
plot(  
  Employee$Years_of_Service,  
  Employee$Performance_Score,  
  type = "o",  
  col = "blue",  
  pch = 16,  
  xlab = "Years of Service",  
  ylab = "Performance Score",  
  main = "Employee Performance Trend"  
)
```

OUTPUT:



2. Bar Chart Showing the Distribution of Employees Across Different Departments

CODE:

```
Employee <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
  
Department_Count <- table(Employee$Department)  
print(Department_Count)  
  
barplot(  
  Department_Count,  
  col = c("skyblue", "lightgreen", "pink"),  
  main = "Employees by Department",  
  xlab = "Department",  
  ylab = "Number of Employees"  
)
```

OUTPUT:



3. Table to Display the Performance Data for Each Employee

CODE:

```
Employee <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
print(Employee)
```

OUTPUT:

```
> print(Employee)  
  Employee_ID Department Years_of_Service Performance_Score  
1           1     Sales              5              85  
2           2       HR              3              92  
3           3  Marketing              7              78  
> |
```

4. Line Chart, Bar Chart, and Table Together (Dashboard in R)

CODE:

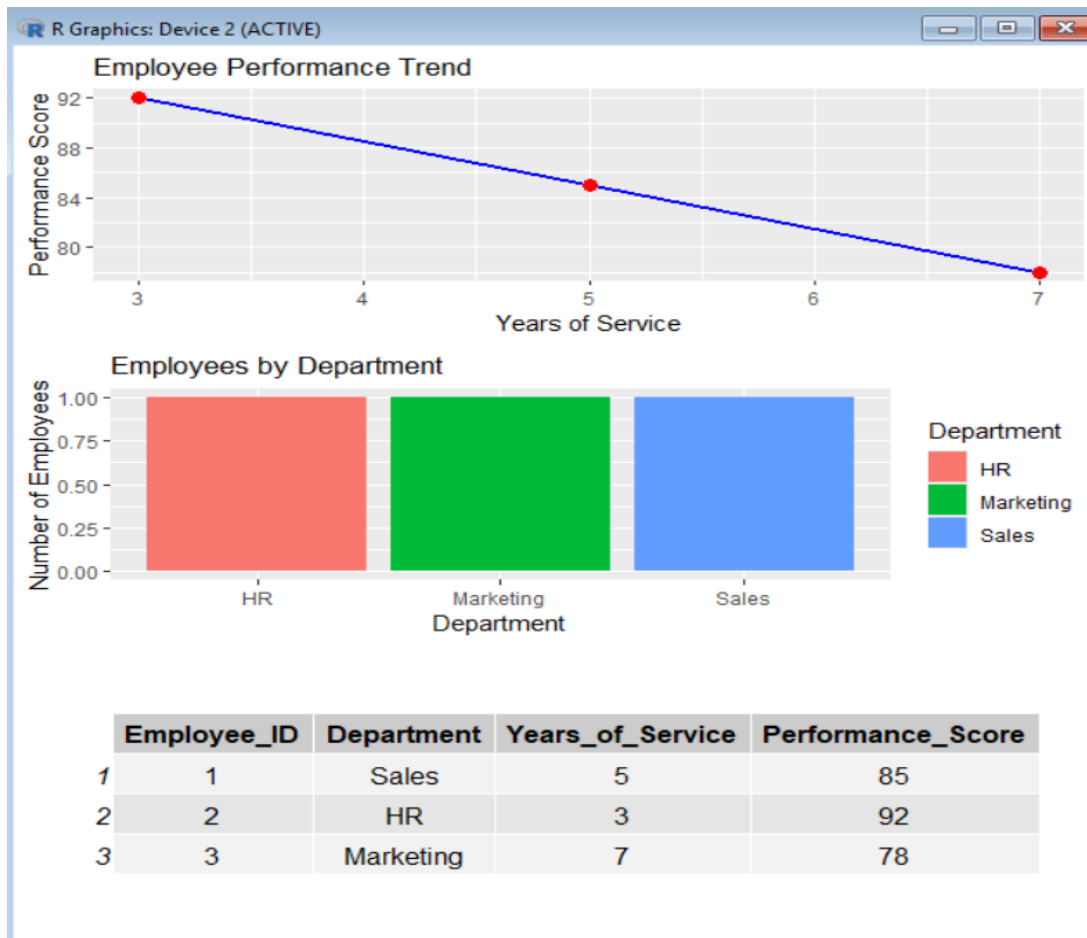
```
library(gridExtra)  
library(grid)  
library(ggplot2)  
Employee <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
Line_Chart <- ggplot(Employee,  
  aes(x = Years_of_Service,  
      y = Performance_Score,  
      group = 1)) +
```

```

geom_line(color = "blue", size = 1) +
geom_point(size = 3, color = "red") +
labs(
  title = "Employee Performance Trend",
  x = "Years of Service",
  y = "Performance Score"
)
Bar_Chart <- ggplot(Employee,
  aes(x = Department,
    fill = Department)) +
geom_bar() +
labs(
  title = "Employees by Department",
  x = "Department",
  y = "Number of Employees"
)
Employee_Table <- tableGrob(Employee)
grid.arrange(
  Line_Chart,
  Bar_Chart,
  Employee_Table,
  ncol = 1
)

```

OUTPUT:



SET 18: PRODUCT INVENTORY MANAGEMENT

1. Bar Chart to Visualize the Quantity of Each Product

CODE:

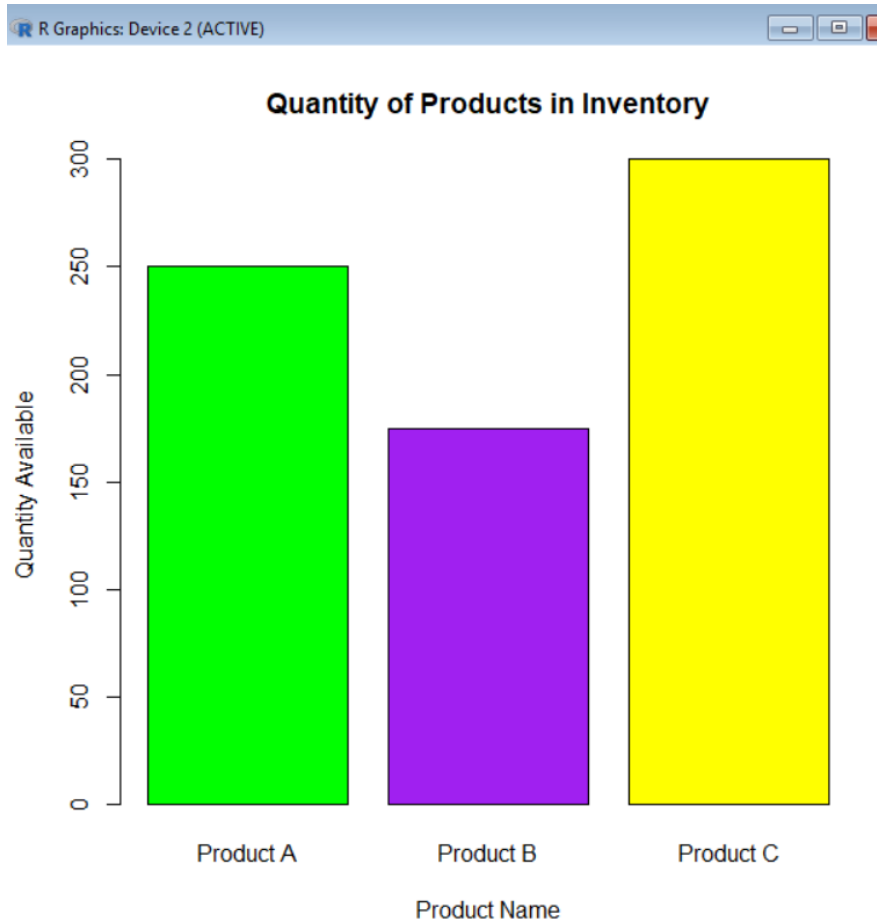
```
Product <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

print(Product)

barplot(
  Product$Quantity_Available,
  names.arg = Product$Product_Name,
  col = c("green", "purple", "yellow"),
```

```
main = "Quantity of Products in Inventory",  
xlab = "Product Name",  
ylab = "Quantity Available"  
)
```

OUTPUT:



2. Stacked Bar Chart to Show the Quantity of Each Product Within Different Product Categories

CODE:

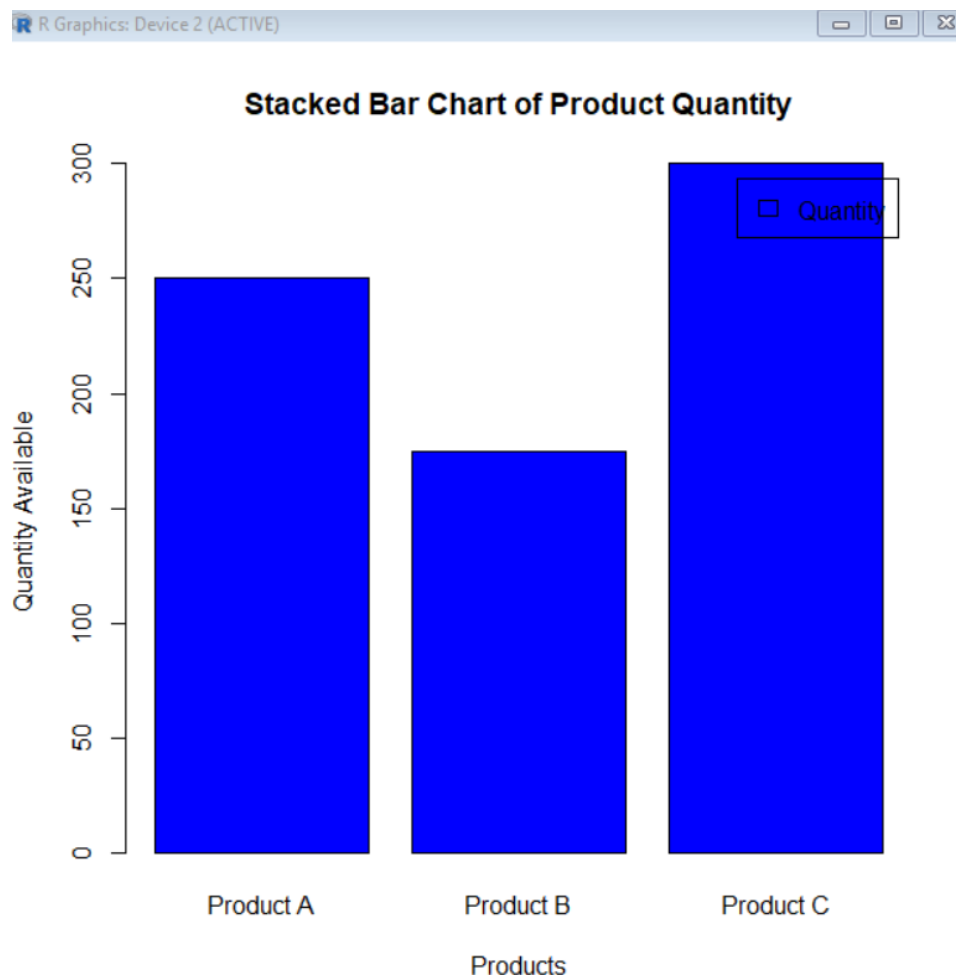
```
Product <- data.frame(  
  Product_Name = c("Product A", "Product B", "Product C"),  
  Quantity_Available = c(250, 175, 300)  
)  
  
Quantity <- matrix(  
  Product$Quantity_Available,
```

```

nrow = 1
)
colnames(Quantity) <- Product$Product_Name
rownames(Quantity) <- "Quantity"
barplot(
  Quantity,
  col = c("blue"),
  main = "Stacked Bar Chart of Product Quantity",
  xlab = "Products",
  ylab = "Quantity Available",
  legend.text = rownames(Quantity)
)

```

OUTPUT:



3. Table to Show the Inventory Data for Each Product

CODE:

```
Product <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Product_Name = c("Product A", "Product B", "Product C"),  
  Quantity_Available = c(250, 175, 300),  
  Price = c(20, 15, 18)  
)  
print(Product)
```

OUTPUT:

```
> print(Product)  
  Product_ID Product_Name Quantity_Available Price  
1          1   Product A             250      20  
2          2   Product B             175      15  
3          3   Product C             300      18  
> |
```

4. Display the Bar Chart, Stacked Bar Chart, and Table Together (Dashboard in R)

CODE:

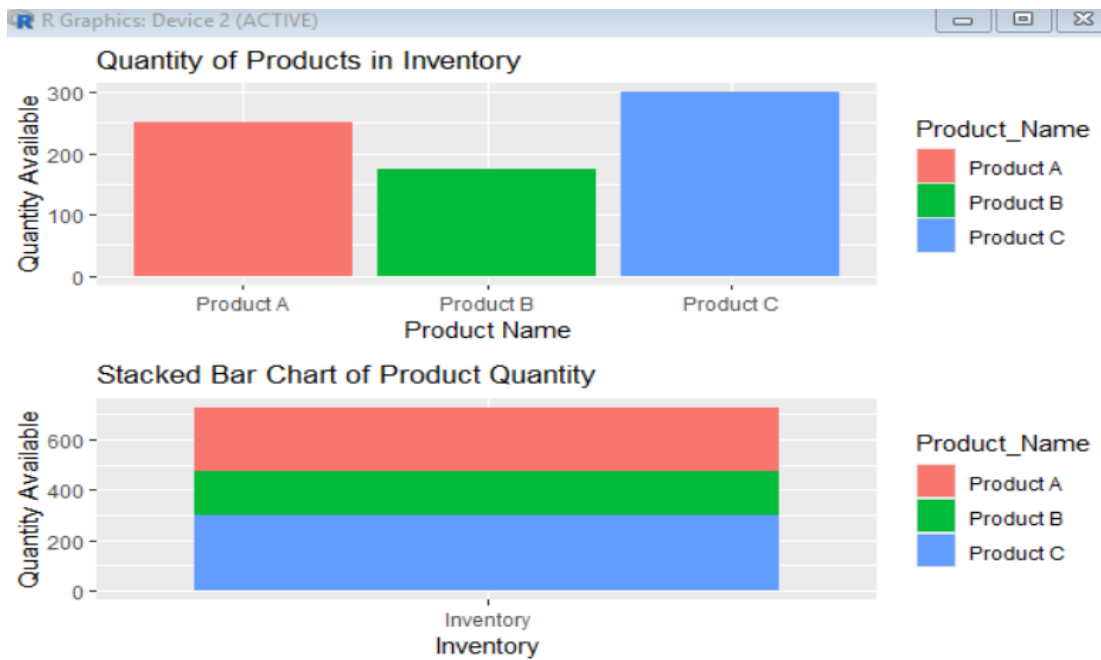
```
library(ggplot2)  
library(gridExtra)  
library(grid)  
Product <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Product_Name = c("Product A", "Product B", "Product C"),  
  Quantity_Available = c(250, 175, 300),  
  Price = c(20, 15, 18)  
)  
Bar_Chart <- ggplot(Product,  
  aes(x = Product_Name,  
    y = Quantity_Available,
```

```

        fill = Product_Name)) +
geom_bar(stat = "identity") +
labs(
  title = "Quantity of Products in Inventory",
  x = "Product Name",
  y = "Quantity Available"
)
Stacked_Bar <- ggplot(Product,
  aes(x = "Inventory",
    y = Quantity_Available,
    fill = Product_Name)) +
geom_bar(stat = "identity") +
labs(
  title = "Stacked Bar Chart of Product Quantity",
  x = "Inventory",
  y = "Quantity Available"
)
Inventory_Table <- tableGrob(Product)
grid.arrange(
  Bar_Chart,
  Stacked_Bar,
  Inventory_Table,
  ncol = 1
)

```

OUTPUT:



	Product_ID	Product_Name	Quantity_Available	Price
1	1	Product A	250	20
2	2	Product B	175	15
3	3	Product C	300	18

SET 19: SURVEY RESPONSES ANALYSIS

1. Grouped Bar Chart to Visualize the Distribution

CODE:

```
Survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

Q1_Count <- table(Survey$Question1)
print(Q1_Count)

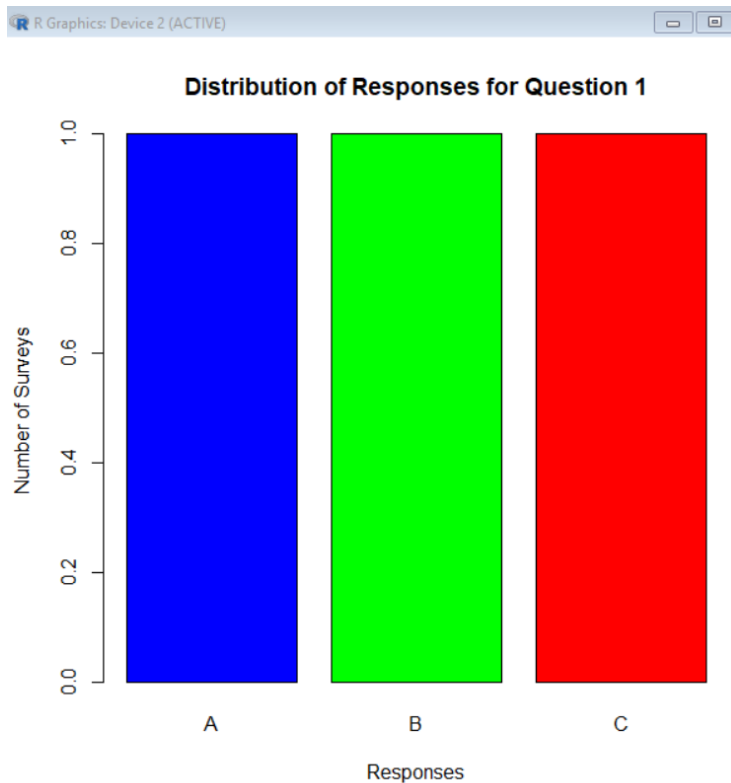
barplot(
  Q1_Count,
  col = c("blue", "green", "red"),
```

```

main = "Distribution of Responses for Question 1",
xlab = "Responses",
ylab = "Number of Surveys",
beside = TRUE
)

```

OUTPUT:



2. Stacked Bar Chart

CODE:

```

Survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

Response_Count <- rbind(
  table(Survey$Question1),

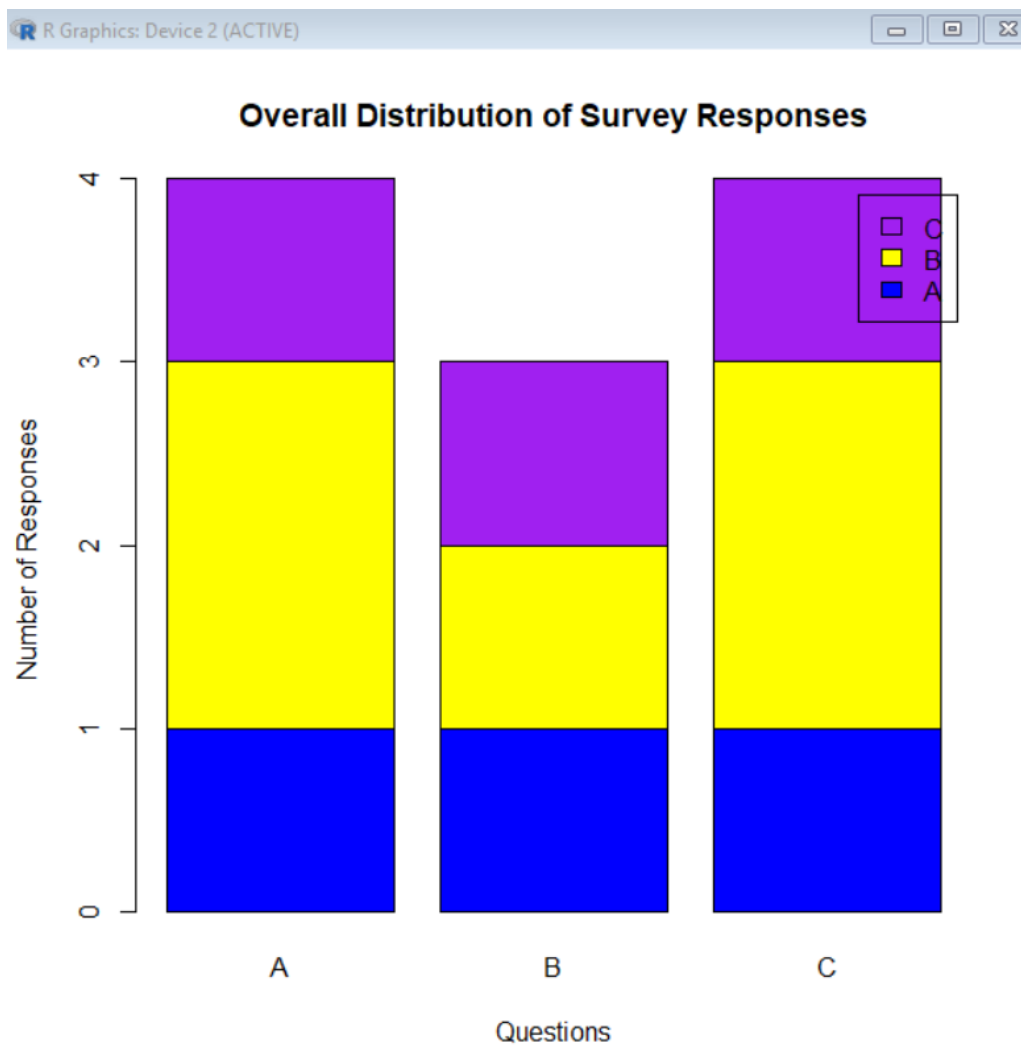
```

```

table(Survey$Question2),
table(Survey$Question3)
)
barplot(
  Response_Count,
  col = c("blue", "yellow", "purple", "orange"),
  main = "Overall Distribution of Survey Responses",
  xlab = "Questions",
  ylab = "Number of Responses",
  legend.text = colnames(Response_Count)
)

```

OUTPUT:



3. Table to Show the Survey

CODE:

```
Survey <- data.frame(  
  Survey_ID = c(1, 2, 3),  
  Question1 = c("A", "B", "C"),  
  Question2 = c("B", "A", "A"),  
  Question3 = c("C", "D", "B")  
)  
print(Survey)
```

OUTPUT:

```
> print(Survey)  
  Survey_ID Question1 Question2 Question3  
1         1         A         B         C  
2         2         B         A         D  
3         3         C         A         B  
> |
```

4. Display the Grouped Bar Chart, Stacked Bar Chart, and Table Together (Dashboard in R)

CODE:

```
library(ggplot2)  
library(gridExtra)  
library(grid)  
Survey <- data.frame(  
  Survey_ID = c(1, 2, 3),  
  Question1 = c("A", "B", "C"),  
  Question2 = c("B", "A", "A"),  
  Question3 = c("C", "D", "B")  
)  
Q1_Count <- as.data.frame(table(Survey$Question1))  
Grouped_Bar <- ggplot(Q1_Count,  
  aes(x = Var1,
```

```

      y = Freq,
      fill = Var1)) +
geom_bar(stat = "identity") +
labs(
  title = "Distribution of Responses for Question 1",
  x = "Responses",
  y = "Count"
)
Survey_Long <- data.frame(
  Question = rep(c("Question1", "Question2", "Question3"), each = 3),
  Response = c(Survey$Question1, Survey$Question2, Survey$Question3)
)

Stacked_Bar <- ggplot(Survey_Long,
  aes(x = Question,
      fill = Response)) +
geom_bar() +
labs(
  title = "Overall Distribution of Survey Responses",
  x = "Questions",
  y = "Count"
)
Survey_Table <- tableGrob(Survey)
grid.arrange(
  Grouped_Bar,
  Stacked_Bar,
  Survey_Table,
  ncol = 1
)

```

OUTPUT:



SET 20: STOCK ANALYSIS

1. Line Chart to Visualize the Stock Prices of Three Companies

CODE:

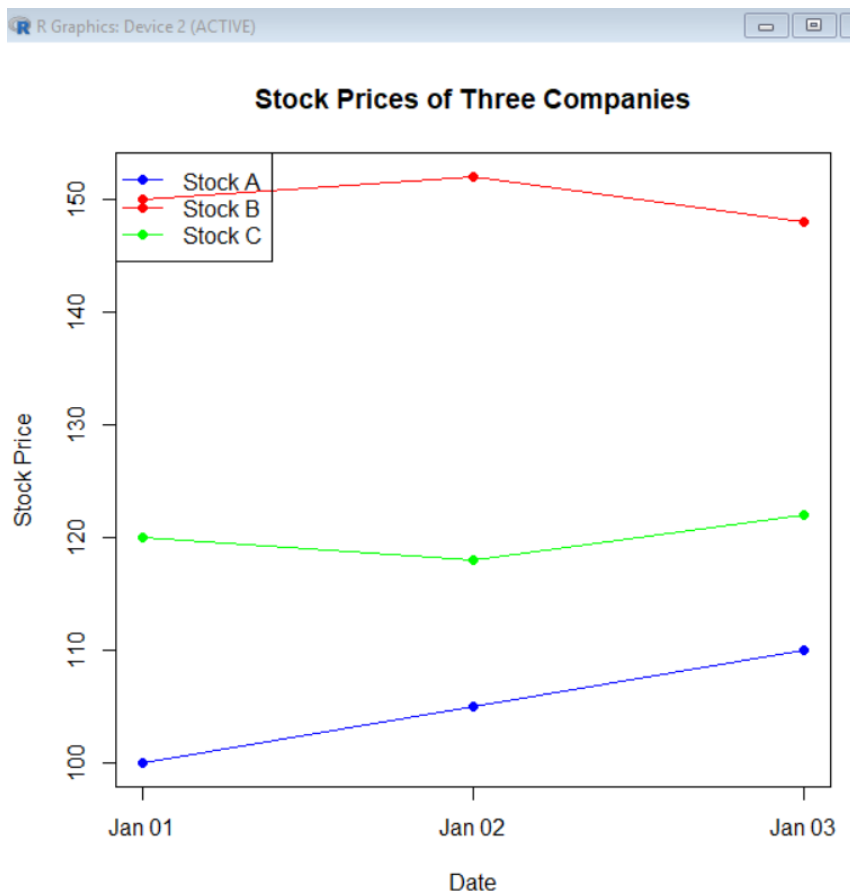
```
Stock <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
  Stock_A = c(100, 105, 110),  
  Stock_B = c(150, 152, 148),  
  Stock_C = c(120, 118, 122)  
)  
print(Stock)  
matplot(  
  Stock$Date,  
  Stock[,2:4],  
  type = "o",
```

```

pch = 16,
lty = 1,
col = c("blue", "red", "green"),
xlab = "Date",
ylab = "Stock Price",
main = "Stock Prices of Three Companies"
)
legend(
  "topleft",
  legend = c("Stock A", "Stock B", "Stock C"),
  col = c("blue", "red", "green"),
  lty = 1,
  pch = 16
)

```

OUTPUT:

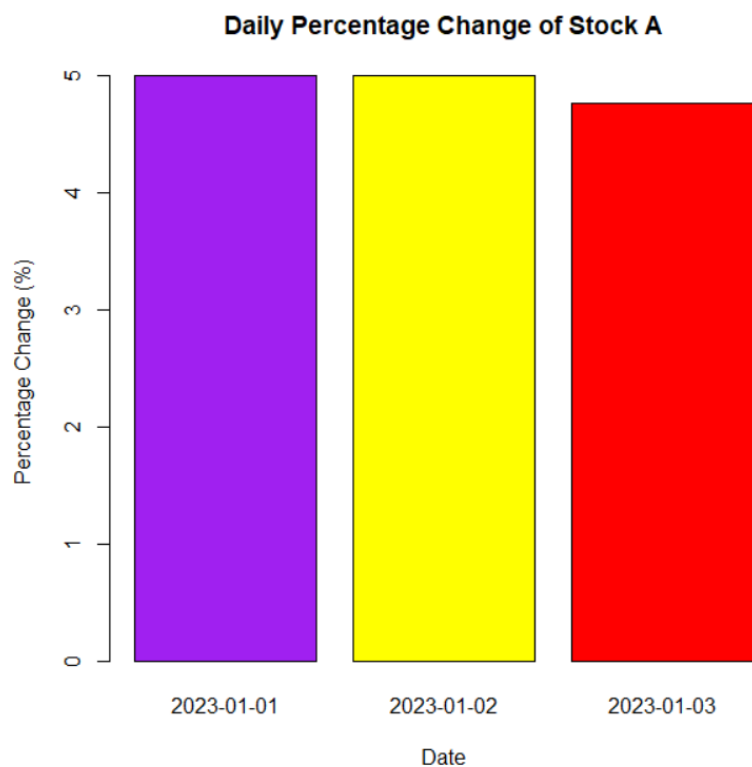


2. Bar Chart Showing the Daily Percentage Change in Stock Prices for Stock A

CODE:

```
Stock <- data.frame(  
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),  
  Stock_A = c(100, 105, 110)  
)  
Percent_Change <- c(  
  5,  
  ((Stock$Stock_A[2]-Stock$Stock_A[1])/Stock$Stock_A[1])*100,  
  ((Stock$Stock_A[3]-Stock$Stock_A[2])/Stock$Stock_A[2])*100  
)  
print(Percent_Change)  
barplot(  
  height = Percent_Change,  
  names.arg = Stock$Date,  
  col = c("purple", "yellow", "red"),  
  main = "Daily Percentage Change of Stock A",  
  xlab = "Date",  
  ylab = "Percentage Change (%)"  
)
```

OUTPUT:



3. Table to Display the Stock Price Data

CODE:

```
Stock <- data.frame(  
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),  
  Stock_A = c(100, 105, 110),  
  Stock_B = c(150, 152, 148),  
  Stock_C = c(120, 118, 122)  
)  
print(Stock)
```

OUTPUT:

```
> print(Stock)  
      Date Stock_A Stock_B Stock_C  
1 2023-01-01     100     150     120  
2 2023-01-02     105     152     118  
3 2023-01-03     110     148     122  
> |
```

4. Display the Line Chart, Bar Chart, and Table Together (Dashboard in R)

CODE:

```
library(ggplot2)
library(gridExtra)
library(grid)

Stock <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

Stock_Long <- data.frame(
  Date = rep(Stock$Date, 3),
  Price = c(Stock$Stock_A, Stock$Stock_B, Stock$Stock_C),
  Company = rep(c("Stock A", "Stock B", "Stock C"), each = 3)
)

Line_Chart <- ggplot(Stock_Long,
  aes(x = Date,
      y = Price,
      color = Company,
      group = Company)) +
  geom_line(size = 1) +
  geom_point(size = 3) +
  labs(
    title = "Stock Prices of Three Companies",
    x = "Date",
    y = "Stock Price"
  )

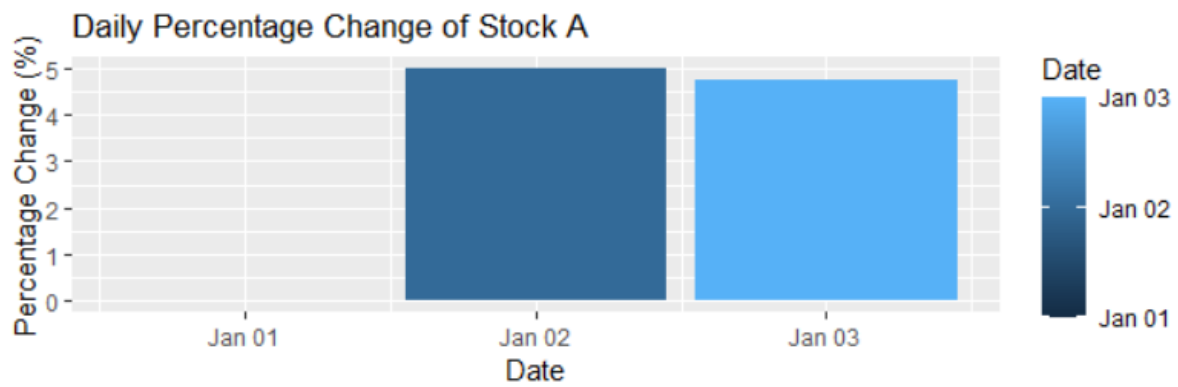
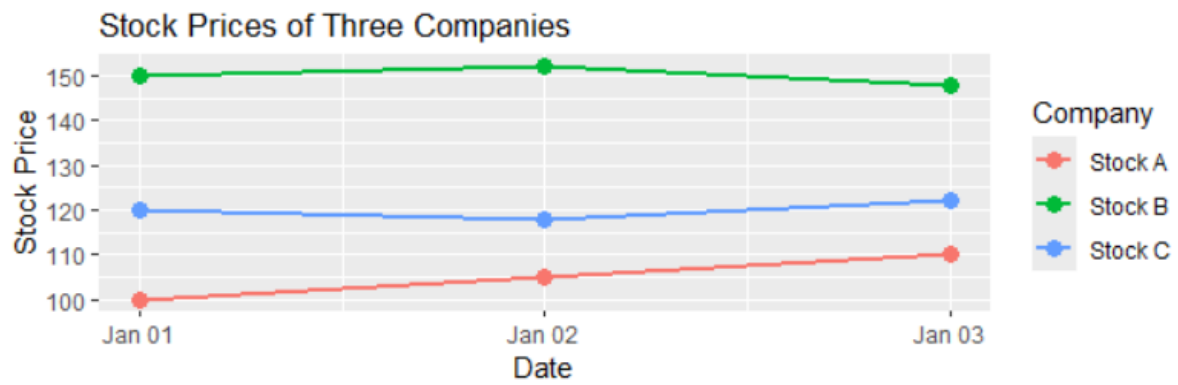
Percent_Change <- c(
```

```

0,
diff(Stock$Stock_A) / head(Stock$Stock_A, -1) * 100
)
Bar_Data <- data.frame(
  Date = Stock$Date,
  Change = Percent_Change
)
Bar_Chart <- ggplot(Bar_Data,
  aes(x = Date,
    y = Change,
    fill = Date)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Daily Percentage Change of Stock A",
    x = "Date",
    y = "Percentage Change (%)"
  )
Stock_Table <- tableGrob(Stock)
grid.arrange(
  Line_Chart,
  Bar_Chart,
  Stock_Table,
  ncol = 1
)

```

OUTPUT:



	Date	Stock_A	Stock_B	Stock_C
1	2023-01-01	100	150	120
2	2023-01-02	105	152	118
3	2023-01-03	110	148	122